The food model, worked example (specification version 3, M98)

The equation numbers and symbols are those of `food_model_spec.md`. All values are computed by `plan/tools/food_model.py` and `plan/tools/food_model_trace.py` from the data files, and are rounded for display.

City E (rich city):

- $N = 100{,}000$;
- $\mathbf{w} = (w_{\text{vig}}, w_{\text{act}}, w_{\text{sed}}) = (0.2,\ 0.4,\ 0.4)$;
- stocks S_g in tonnes: Cow 20,000; Dairy 150,000; Potato 60,000; Wheat 40,000; Apple 20,000; Sugar 8,000; Wine 15,000; Tea 400; Salt 500.

M1. Availability, equation (1)

$$a_g = \frac{S_g \cdot 10^6}{365 \cdot 100{,}000}$$

g	Cow	Dairy	Potato	Wheat	Apple	Sugar	Wine	Tea	Salt
a_g [g/(p·d)]	547.9	4,109.6	1,643.8	1,095.9	547.9	219.2	411.0	11.0	13.7

M2. Need and bounds, equations (2)-(8)

(2) tea: $\ell\chi/1000 = 43 \cdot 200/1000 = 8.6$ mg/g; $\quad q_{\text{tea}} = \min(11.0,\ 600/8.6) = 11.0$ g
caffeine $= 11.0 \cdot 8.6 = 94.2$ mg; $\quad c = 94.2/600 = 0.157$

(3) $\Lambda^{\text{w}} = 0.2 \cdot 2.25 + 0.4 \cdot 1.76 + 0.4 \cdot 1.53 = 1.766$

(4) $E^{\text{need}} = 1536 \cdot 1.766 + 150 \cdot 0.157 = 2{,}735$ kcal

(5) $E^{\max} = 1536 \cdot 2.40 + 150 \cdot 0.157 = 3{,}709$ kcal

(6) $F^{\text{fed}} = 1.36 \cdot 2{,}735 = 3{,}720$; $\quad F^{\max} = 1.36 \cdot 3{,}709 = 5{,}044$

(7) $P^{\max} = 0.35 \cdot 3{,}709/4 = 324.5$ g

(8) $q_{\text{salt}} = \min(13.7,\ 10) = 10$ g

M3. Indicators per good, equations (9)-(16)

Inputs from the data:

g	ε_g	π_g	φ_g	η_g	δ_g	**Holt** H_g
Cow (composite, 28)	150	18.5	7.9	1.0	1.12	176
Dairy	61	3.3	3.3	1.0	1.14	– (formula: 155)
Potato	67	1.6	0.1	1.0	0.44	323
Wheat	334	12.2	2.3	0.8	0.44	157 (wholemeal bread)
Apple	48	0.1	0.3	1.0	0.44	197

g	ε_g	π_g	φ_g	η_g	δ_g	**Holt** H_g
Sugar	373	0	0	1.0	0.44	– (formula: 100)
Wine	68	0	0	1.0	0.44	– (formula: 140)

Results of (9)-(16):

g	e_g **(9)**	p_g **(10)**	σ_g **(11)**	s_g **(12)**	ν_g **(13)**	β_g **(14)**	$K_g^{(1)}$ **(15)**	$K_g^{(2)}$ **(15)**	$\bar{q}_g$ **(16)**
Cow	1.500	0.493	1.76	1.760	0.710	−0.139	2.470	−0.139	548
Dairy	0.610	0.216	0.78	1.270	0.795	0.858	2.065	0.858	3,000
Potato	0.670	0.096	3.23	0.961	0.659	0.365	1.620	0.365	1,644
Wheat	2.672	0.146	1.57	1.000	0.551	0.133	1.551	0.133	1,096
Apple	0.480	0.008	1.97	0.893	0.462	1.491	1.355	1.491	548
Sugar	3.730	0.000	1.00	0.886	0.173	1.071	1.059	1.071	219
Wine	0.680	0.000	0.70	0.886	0.292	1.467	1.178	1.467	411

Units: e_g in kcal/g; $\bar{q}_g$ in g/(p·d); the rest are dimensionless.

One row in full: Cow.

(9) $e = 150 \cdot 1.0/100 = 1.500; \qquad \hat{\varphi} = 7.9/100 = 0.079$

(10) $p = 4 \cdot 18.5/150 = 0.493; \quad f = 9 \cdot 7.9/150 = 0.474; \quad k = 1 - 0.474 - 0.493 = 0.033; \quad z = l = 0$

(11) $\sigma = 176/100 = 1.76$

(12) $s = \dfrac{1 + 2.0 \cdot 0.493 \cdot 1}{1 + 2.0 \cdot 0.146 \cdot 0.44} = \dfrac{1.986}{1.128} = 1.760$

(13) $\nu = \frac{1}{13}\big(\underbrace{1}_{\text{prot}} + \underbrace{0}_{\text{C}} + \underbrace{1}_{\text{A}} + \underbrace{0.97}_{\text{B1}} + \underbrace{1}_{\text{B3}} + \underbrace{0.56}_{\text{fol}} + \underbrace{1}_{\text{B12}} + \underbrace{1}_{\text{Fe}} + \underbrace{1}_{\text{Zn}} + \underbrace{0.10}_{\text{Ca}} + \underbrace{0}_{\text{I}} + \underbrace{0.70}_{\text{K}} + \underbrace{0.90}_{\text{Na}}\big) = \frac{9.226}{13} = 0.710$

(14) $\beta = 1 \cdot 0 + 2 \cdot 0.474 \cdot 0.033 + 1 \cdot 0 + 1 \cdot \min(1, 0.201/3) - 0.5 \cdot 0.474 = 0.031 + 0.067 - 0.237 = -0.139$

(15) $K^{(1)} = 1.760 + 0.710 = 2.470; \qquad K^{(2)} = -0.139$

(16) $\bar{q} = \min(547.9,\ 250/0.079,\ \cdot) = \min(547.9,\ 3165) = 547.9$

M4. Allocation, Algorithm 1 with (17)-(18)

Pass	Key	Stops at	Ends with
1	$K^{(1)}$ (nutrition)	$F \geq F^{\text{fed}} = 3{,}720$	$E = 2{,}120$ kcal, $F = 3{,}735$
2	$K^{(2)}$ (well-being; only $K^{(2)} > 0$, so Cow is out)	$F \geq F^{\max} = 5{,}044$	$E = 3{,}300$ kcal, $F = 5{,}062$

Result:

g	q_g **[g]**	$e_g q_g$ **[kcal]**	$\sigma_g e_g q_g$ **[fullness]**	**taken in pass**
Cow	333	500	880	1
Dairy	1,934	1,180	920	1, 2
Potato	851	570	1,841	1, 2
Wheat	135	360	565	1
Apple	542	260	512	1, 2 (stock)
Sugar	43	160	160	1, 2

g	q_g [g]	$e_g q_g$ [kcal]	$\sigma_g e_g q_g$ [fullness]	taken in pass
Wine	397	270	189	1, 2 (stock)
Σ		**3,300**	**5,062**	

A good can appear in both passes because (17) declines with q_g. Pass 2 continues the good from where pass 1 left it.

Check of property (P1), equation (29): $\tilde{\sigma} = 5{,}062/3{,}300 = 1.534$, so $E = F/\tilde{\sigma} = 3{,}300$ kcal. The stop overshoots $F^{\max}$ by 18, within one step (P3).

M5. Outcomes, equations (19)-(27)

$$(19)\quad fed = 5{,}062/3{,}720 = 1.36; \qquad x = 3{,}300/2{,}735 = 1.21$$

$$(21)\quad a^{\mathrm{Fe}} = 0.10 + 0.08 \cdot (\text{animal protein}/P) = 0.166; \qquad P = 152.8\ \text{g}\ (18.5\,\%\ \text{of energy})$$

$$(23)\quad W^{\text{food}} = 1{,}325\ \text{ml}; \qquad W^{\text{drink}} = \underbrace{1934 \cdot 1.5}_{\text{milk}} + \underbrace{397}_{\text{wine}} + \underbrace{11.0 \cdot 43}_{\text{tea}} = 3{,}770\ \text{ml}$$

$$(24)\quad W^{\text{plain}} = \max(0,\ 2{,}250 - 1{,}325 - 3{,}770) = 0$$

$$(25)\quad \Lambda^{\text{act}} = \min(2.40, \max(1.40,\ 3{,}300/1{,}536)) = 2.149$$

$$(26)\quad work = \min(1.20,\ (2.149 - 1.40)/(1.766 - 1.40)) = \min(1.20,\ 2.05) = 1.20$$

Coverage, (20) and (22):

i	C	A	B1	B3	folate	B12	Fe	Zn	Ca	I	K	Na
c_i	1.82	2.46	3.37	2.42	1.18	5.82	2.76	2.73	3.18	2.12	2.36	8.30

Stocks, (27), in tonnes:

g	Cow	Dairy	Potato	Wheat	Apple	Sugar	Wine	Tea	Salt
L_g	7,833	79,393	28,948	35,082	229	6,434	507	0	135
D_g	7,833	79,393	14,474	1,754	229	643	76	0	14

Fresh goods ($\bar{\delta} = 1$) spoil whole; grain, sugar and wine keep. Milk converted to cheese before eating (recipes) is not in this example.